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Portuguese Language Student Performance Part 3

This analysis examines the factors that influence student academic performance, with a focus on predicting final grades (G3). Using regression modeling, the goal was to identify which variables most strongly drive student outcomes and to evaluate how well these variables collectively explain variation in grades. The model achieved a high level of explanatory power ($R^2 \approx 0.86$), indicating that the selected features capture most of the important influences on student performance.

The results show that prior academic performance is the strongest predictor of final grades by far. In particular, earlier period grades (G1 and G2) have the largest positive coefficients, meaning students who perform well earlier in the course are highly likely to perform well at the end. This suggests strong grade persistence over time. While other important drivers are past class failures, study time, absences and parental and social factors. Study time is a positive relationship while failures and absences are negative relationships. Some variables had minimal or negligible impact, suggesting that not all collected features are equally useful predictors. Overall, academic habits and prior achievement dominate the model's predictive power.

A key takeaway is that early academic performance strongly determines final outcomes. So, one effective policy change would be to implement an early intervention system based on first-period grades (G1). Schools could identify students who are underperforming early in the

term and provide targeted support such as tutoring, mentoring, or structured study programs.

Because the model shows that early grades are highly predictive, intervening early can significantly improve final outcomes. This approach is more effective than waiting until the end of the course, when it is often too late to reverse poor performance. Focusing on early detection and support can help break negative academic patterns and improve overall student success.